Object Detection and Multi Object Tracking on MOT17 Benchmark

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https://github.com/Sreebhargavibalijaa/CSE8001---Object-Detection

Introduction

The project develops one end to end pedestrian detection and multi-object tracking pipeline using the MOT17 benchmark. The objective of Milestone 1 was to build multiple functional pipeline configurations and compare their detection and tracking quality and latency. The pipeline combines three pretrained object detectors, YOLO26n, Faster RCNN resnet50-FPN, and DETR resnet-50 with two trackers bytetrack and oc-sort resulting in six configurations. The main challenge we see is that detecting a person in one frame is not enough, the system must also preserve the same identity across consecutive frames despite occlusion, crowding, motion. This creates a trade off between tracking quality and computational speed.

Related Work

Modern multi-object tracking commonly follows a tracking-by-detection approach. Bytetrack improves association by also using lower-confidence detections, which helps recover partially occluded pedestrians. OC-SORT focuses on improving motion consistency when detections disappear temporarily or movement becomes non-linear. Bot-SORT extends tracking further using appearance information, while centertrack combines detection and tracking more closely. Motchallenge provides the standardized MOT17 benchmark and evaluation metrics used in this project.

Dataset and Pipeline Architecture

The MOT17 dataset contains pedestrian video sequences recorded in different crowd and motion conditions. Each sequence is stored as a set of ordered image frames along with information such as image size and ground truth identities for the training data. Before running the models, we read the frames using OpenCV and apply the preprocessing required by each detector, mainly resizing and normalization. We then keep only pedestrian detections and convert the bounding boxes into a common format for tracking. For Milestone 1, we tested all six detector-tracker combinations on two complete sequences: MOT17-02-DPM**,** with 600 frames and about 31 pedestrians per frame and MOT17-09-DPM with 525 frames and about 10 pedestrians per frame.

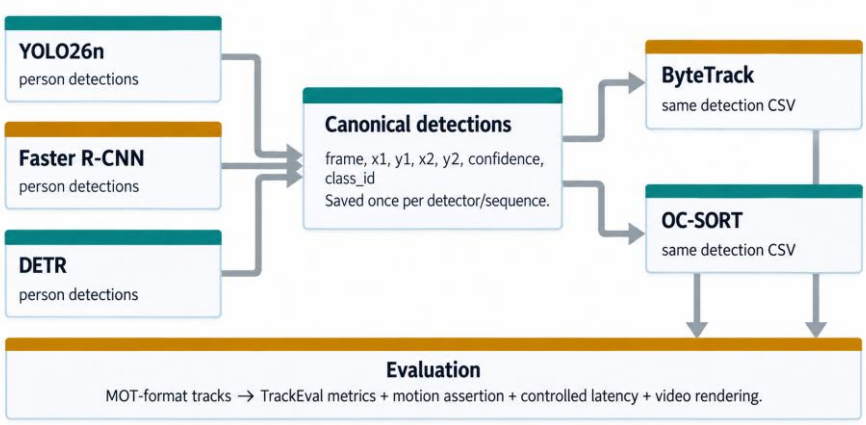


Figure-1 Pipeline

The pipeline follows a tracking by detection architecture. Each video frame is first processed by one of three object detectors, which were YOLO26n, Faster R-CNN ResNet50-FPN or DETR ResNet-50 and the person detections are saved in a common format containing the frame number, bounding-box coordinates, confidence score and class ID. These saved detections are then passed to either ByteTrack or OC-SORT**,** producing six total detector-tracker configurations. The resulting tracks are stored in MOT format and evaluated using TrackEval metrics, a motion-continuity assertion, latency measurements and tracking-video visualizations.

SLA

For Milestone 1, all latency and throughput measurements were collected on a local CPU without GPU acceleration or deployment specific optimization. Since SLA performance depends strongly on the machine configuration, we report the observed FPS and latency as baseline system measurements for this hardware setup. These measurements are used to compare the six detector–tracker configurations under the same conditions and to identify the trade-off between tracking quality and computational performance.

Evaluation and Results

The pipelines are evaluated using standard motchallenge metrics and system level measurements.

Metric	Meaning
HOTA	Balances detection and association quality
MOTA	Penalizes false negatives, false positives, and ID switches
IDF1	Measures identity consistency over time
Precision	Fraction of predicted pedestrians that are correct
Recall	Fraction of real pedestrians detected
FPS	Processing speed
Motion assertion	Checks for unrealistic movement of the same track

Preliminary Baseline Results

Before the final six-configuration comparison, an initial evaluation was carried out. YOLO26n with ByteTrack was compared against the public MOT17 detector baselines using a simple IoU tracker. These results were mainly used as an early reference point for understanding the effect of detector quality and tracking method.

Pipeline	HOTA	MOTA	IDF1	Precision	Recall	IDSW
YOLO26n + ByteTrack	26.13	14.53	22.49	97.78	14.91	8
DPM + IoU tracker	14.35	10.00	15.65	65.93	25.78	456
Faster R-CNN + IoU tracker	30.90	25.74	34.57	80.21	35.34	164
SDP + IoU tracker	25.89	29.57	30.52	76.95	48.20	778

For the final experiment, all three detectors were paired with both ByteTrack and OC-SORT, giving six complete configurations. The same saved detections were used for both trackers to make the comparison fair.

Detector	Tracker	HOTA	MOTA	IDF1	Precision	Recall	FPS
YOLO26n	Bytetrack	31.12	26.95	34.39	89.80	30.93	4.17
YOLO26n	OC-SORT	31.87	26.86	36.07	89.52	30.97	4.15
Faster R-CNN	Bytetrack	31.72	18.14	36.04	61.21	54.43	0.13
Faster R-CNN	OC-SORT	30.68	17.36	35.62	60.75	54.55	0.13
DETR	Bytetrack	28.02	4.80	28.87	54.37	53.15	0.24
DETR	OC-SORT	26.69	2.15	26.66	53.33	52.86	0.24

YOLO26n + OC-SORT produced the best balanced tracking quality, with the highest HOTA of 31.87 and IDF1 of 36.07. YOLO26n + bytetrack produced the highest MOTA of 26.95 and the fastest measured throughput of 4.17 FPS. Faster R-CNN + bytetrack achieved similar HOTA and IDF1 to the best YOLO configurations, but ran at only 0.13 FPS, making it about 32 times slower than YOLO + bytetrack in the controlled CPU experiment. DETR produced lower tracking scores under the current pretrained, CPU-only setup.

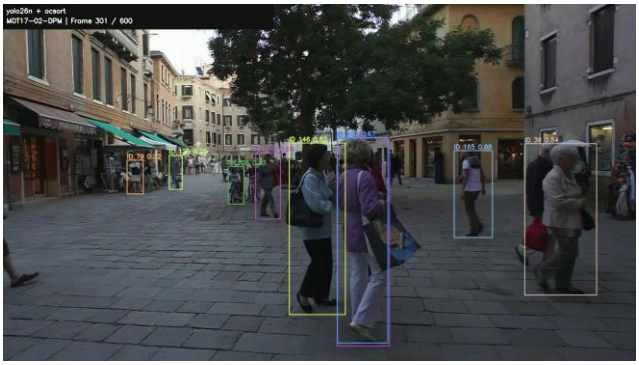


Figure 2 Dense Scene MOT17-02-DPM



Figure-3 Lower Density MOT17-09-DPM

Performance also changed strongly with scene density. All six configurations achieved higher HOTA on MOT17-09 than MOT17-02, showing that crowded scenes are significantly more difficult and that input distribution affects pipeline performance.

LIMITATIONS

The final evaluation uses two MOT17 training sequences rather than the full training split. All detectors use generic pretrained weights without MOT-specific fine-tuning, thresholds were not optimized for each configuration, and latency was measured on CPU rather than optimized GPU hardware.

TEAM ORGANIZATION

We equally worked on the milestone1 using a shared GitHub repository for collaboration. We divided the work based on the two main stages of the pipeline. Sree handled the initial frame processing and tested the three object detection models and Preya then extended the pipeline by adding ByteTrack and OC-SORT for object tracking and evaluated the detector-tracker combinations on the two selected MOT17 sequences. Both of us contributed to the literature review, presentation and final repository updates.

CONCLUSION

Overall, Milestone1 helped us successfully produce and evaluate six functional end to end detector-tracker configurations. The results show that model selection cannot be based on tracking accuracy alone. YOLO26n+OC-SORT provides the best balanced quality, while YOLO26n+bytetrack provides the best operational trade off between MOTA and speed. This test also demonstrates that scene distribution has a major effect on tracking performance, providing a useful baseline for later milestones involving deployment, optimization, monitoring, and drift analysis.